

HURON, SD, USAWMO: **726540**Lat: **44.383N**Lon: **98.217W**Elev: **1289**StdP: **14.02**Time Zone: **-6.00 (NAC)**Period: **94-19**WBAN: **14936****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-13.3	-8.2	-19.6	2.0	-12.8	-14.5	2.6	-7.6	34.8	11.4	31.4	15.3	8.8	320	0.644

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.7	93.4	74.0	90.0	72.8	87.0	71.3	77.4	87.9	75.5	85.9	73.8	83.8	12.3	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
74.4	134.9	84.0	72.3	125.5	82.1	70.3	117.2	80.5	41.9	87.7	40.0	86.2	38.2	83.9	83.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 28.2	24.9	21.5	DB	-22.3	99.9	7.4	4.4	-27.6	103.0	-32.0	105.6	-36.1	108.1	-41.5	111.2	(4)
(5)			WB	-22.6	80.2	7.2	1.8	-27.8	81.6	-32.0	82.6	-36.0	83.7	-41.3	85.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	46.2	15.4	20.3	32.7	45.9	58.0	68.4	74.2	71.6	63.3	48.5	33.6	20.7		
	DBStd	22.84	13.24	13.63	13.48	10.37	8.69	6.90	6.40	6.02	8.90	9.76	11.30	12.15		
	HDD50	4237	1073	830	552	194	26	0	0	0	7	147	501	907		
	HDD65	7648	1538	1250	1002	576	248	42	8	14	137	518	943	1372		
	CDD50	2847	0	0	16	71	276	553	751	668	405	99	8	0		
	CDD65	784	0	0	1	3	33	145	293	217	86	6	0	0		
	CDH74	8345	0	0	11	77	427	1509	3179	2119	928	93	2	0		
	CDH80	3166	0	0	2	16	124	509	1369	792	332	22	0	0		
	WSAvg	10.3	10.5	10.8	11.3	11.9	11.0	9.6	8.9	8.7	9.7	10.4	10.4	10.3		
	PrecAvg	21.4	0.5	0.7	1.4	2.3	3.0	3.6	2.7	2.2	2.0	1.7	0.8	0.6		
(15) Precipitation	PrecMax	31.7	2.5	3.9	5.9	6.7	7.7	11.5	6.7	6.5	6.7	5.4	3.9	1.7		
	PrecMin	11.0	0.0	0.0	0.2	0.2	0.3	0.8	0.2	0.2	0.0	0.1	0.0	0.0		
	PrecStd	5.1	0.4	0.6	1.1	1.5	1.8	2.1	1.7	1.3	1.4	1.4	0.8	0.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	49.8	57.4	73.5	81.8	89.4	93.5	99.6	96.2	93.8	83.2	69.4	52.7		
		MCWB	41.5	47.2	54.8	60.2	67.9	73.5	75.7	75.0	70.5	63.2	53.9	42.9		
	2%	DB	41.0	49.3	65.3	75.0	83.3	89.0	94.4	91.2	88.1	75.5	61.6	45.7		
		MCWB	35.9	41.4	52.7	57.0	65.1	72.4	75.4	73.3	69.6	60.4	49.7	38.3		
	5%	DB	37.0	43.1	58.3	69.2	78.7	85.7	90.8	87.5	83.6	70.0	55.0	40.2		
		MCWB	33.8	38.0	48.1	53.7	63.3	70.5	74.2	72.2	67.3	57.2	45.7	35.3		
	10%	DB	33.9	38.1	51.8	64.0	74.2	82.5	87.5	84.3	79.2	64.4	49.7	36.0		
		MCWB	31.6	34.6	45.1	51.0	60.7	68.5	72.3	70.7	65.0	53.9	42.1	32.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	41.6	49.0	58.9	62.7	71.1	77.5	80.5	79.0	74.2	66.7	55.5	44.4		
		MCDB	49.7	58.0	67.6	77.6	83.9	89.2	91.7	89.4	85.6	77.4	66.0	52.1		
	2%	WB	36.6	41.9	53.9	58.6	67.9	75.0	78.3	76.4	72.2	62.0	50.6	39.0		
		MCDB	40.3	48.2	63.7	71.7	79.6	86.2	89.3	86.2	84.2	71.3	60.1	45.1		
	5%	WB	34.0	38.2	49.3	55.1	65.2	72.7	76.5	74.5	70.0	58.6	46.2	35.6		
		MCDB	36.9	42.9	58.1	66.5	76.2	82.9	87.0	84.2	80.6	68.1	54.5	39.4		
	10%	WB	31.6	34.7	44.2	52.3	62.3	70.4	74.5	72.6	67.7	55.0	42.5	33.0		
		MCDB	34.0	37.9	51.7	62.4	72.2	80.0	84.7	81.8	76.4	63.4	49.2	35.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	18.3	19.4	20.0	22.8	22.2	21.7	22.7	22.6	24.5	22.7	20.5	17.5		
		MCDBR	22.2	24.7	31.0	32.3	28.9	24.9	25.6	26.1	28.8	31.0	29.0	23.8		
		MCWBR	18.0	17.6	18.8	17.3	14.8	12.1	10.9	11.6	13.3	17.3	18.1	17.6		
	5% WB	MCDBR	21.8	23.8	29.8	29.0	25.0	22.0	23.1	22.5	24.5	26.9	28.5	22.3		
		MCWBR	18.0	17.2	18.5	16.2	14.0	12.0	11.4	11.4	12.9	16.1	18.2	16.9		
(40) Clear-Sky Solar Irradiance	taub		0.265	0.290	0.312	0.360	0.391	0.381	0.391	0.393	0.348	0.316	0.285	0.266		
	taud		2.450	2.395	2.431	2.307	2.298	2.382	2.364	2.341	2.446	2.500	2.494	2.458		
	Ebn at Noon		277	287	294	286	279	282	277	272	278	274	267	265		
	Edh at Noon		22	28	31	39	41	37	38	37	31	25	21	20		
(44) All-Sky Solar Radiation	RadAvg		584	915	1219	1546	1822	2042	2151	1786	1411	932	629	456		
	RadStd		51	78	113	133	149	144	62	141	95	113	63	36		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)			
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (23 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page